

Predictive stability and adaptation under distribution shift

Calibration, discrimination and decision thresholds in credit-risk modeling

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BENCHMARK → SHIFT → ADAPTATION

START HERE: establish internal performance, then test stability under controlled shifts

01 DATA BENCHMARK

Kaggle loan-default dataset
255,347 records · 29,653 defaults (11.6%)
16 characteristics + income ratio; LoanID excluded.

02 DEVELOPMENT

Train 60% · calibrate 15%
Validate 10% · locked test 15%
Training-only preprocessing; empirical seed 42.

03 MODEL SELECTION

Logistic, ridge logistic, random forest and gradient boosting.
Validation log loss selected raw logistic regression.

04 LOCKED-TEST RESULTS

AUC 0.7585 · PR-AUC 0.3288
Brier 0.0908 · log loss 0.3111
Calibration intercept 0.0073; slope 0.9989.

NEXT: internal validation does not establish deployment stability. The simulations below isolate what changes and which response helps.

1 DISTRIBUTION SHIFT AND DESIGN

P: development distribution · Q: deployment distribution

$P(X, Y) \neq Q(X, Y)$

Covariate shift: population changes; risk mechanism stays fixed.

$P(X) \neq Q(X), \quad P(Y \mid X) = Q(Y \mid X)$

Conditional shift: predictor–outcome relationship changes.

$P(Y \mid X) \neq Q(Y \mid X)$

Covariate shift + misspecification

$X_{1, \text{deploy}} \sim N(\delta, 1), \quad \delta \in \{0, 0.5, 1, 1.5, 2\}$

$\text{logit}(p) = \beta_0 + 0.8X_1 - 0.6X_2 + 0.4X_3 + \beta_4X_1^2$

$\beta_4 \in \{0, 0.25, 0.50\}$

Fitted logistic models omit the quadratic term. Intercepts target approximately 12% development risk.

Intercept shift

$\text{logit}(p_{\text{deploy}}) = \beta_0 + \gamma + 0.8X_1 - 0.6X_2 + 0.4X_3$

Predictor-effect shift

$\beta_{1, \text{deploy}} = 0.8 - \kappa, \quad \kappa \in \{0, 0.4, 0.8, 1.2, 1.6\}$

Deployment intercept adjusted to keep mean risk near 12%.

PROBABILITY ACCURACY



Lower log loss is better. Recalibration nearly matches refitting under intercept shift.

500 replications; seed 1998; 5,000 observations per training, adaptation and test sample. Separate threshold-tuning data.

2 RANKING AND MODEL REPAIR

Calibration is assessed using the original log-odds score:

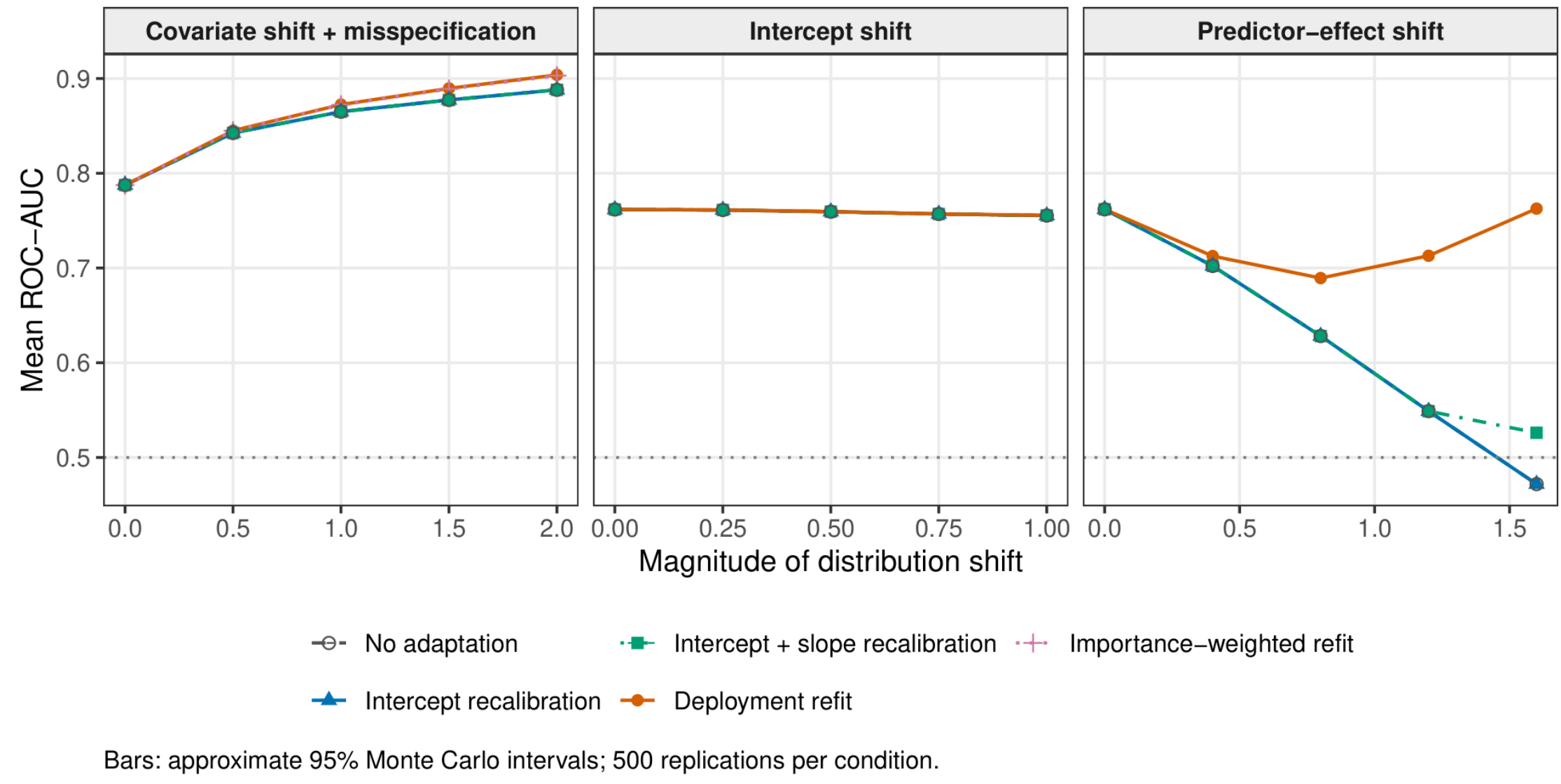
$\text{logit}\{\Pr(Y = 1 \mid \hat{p})\} = a + b \text{logit}(\hat{p})$

For an unchanged true baseline score under intercept shift:

$\text{logit}(p_{\text{deploy}}) = \gamma + \text{logit}(p_{\text{baseline}})$

$a = \gamma, \quad b = 1$

This is the population relationship; finite-sample estimates vary.



Under severe predictor-effect shift, ROC-AUC rises from 0.4719 without adaptation to 0.7627 after refitting.

STRONGEST-SHIFT COMPARISON

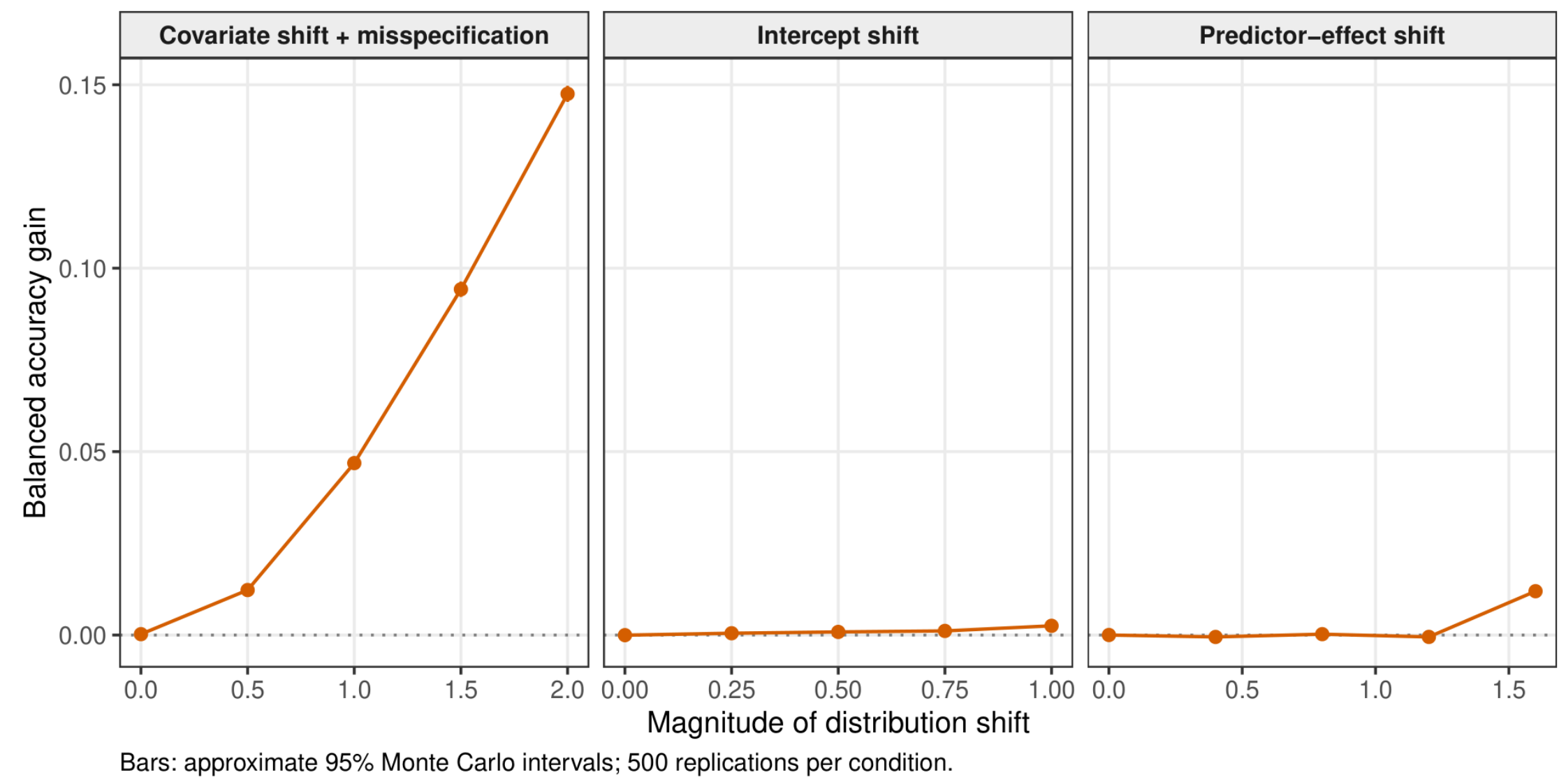
Shift / response	Log loss	Brier	AUC
Covariate + misspecification			
No adaptation	0.5059	0.1676	0.8881
Deployment refit	0.3873	0.1238	0.9039
Intercept shift			
No adaptation	0.5420	0.1743	0.7554
Intercept recalibration	0.4747	0.1545	0.7554
Predictor-effect shift			
No adaptation	0.4366	0.1201	0.4719
Deployment refit	0.3167	0.0933	0.7627

Lower log loss and Brier are better; higher AUC is better. Entries are mean adaptation results over 500 replications. Differences need not imply economic benefit.

3 DECISIONS AND THRESHOLDS

The decision rule changes; predicted probabilities stay fixed.

Balanced accuracy = $\frac{\text{Recall} + \text{Specificity}}{2}$



+14.75 percentage points

Balanced accuracy: 0.6555 → 0.8031

Cutoff: 0.1598 → 0.3940

MATCH THE RESPONSE TO THE CHANGE

Baseline risk changes

Consider intercept recalibration.

Predictor effects change

Consider refitting or redevelopment.

Operating point becomes unsuitable

Reassess the threshold on separate tuning data.

Interpretation limits

The framework is descriptive, not a unique diagnostic rule. Simulated shifts are simplified. Balanced accuracy gives sensitivity and specificity equal weight; lending costs were not optimized.

Probability repair ≠ ranking repair ≠ decision-rule repair

END: good internal performance does not guarantee stability. Match repair to the mechanism and available deployment data.